

```
void setup() {
  oled.begin();

  Serial.begin(9600);

  attachInterrupt(INT0, Meter1ISR, RISING);
  attachInterrupt(INT1, Meter2ISR, RISING);

  Meter1.reset();
  Meter2.reset();
}
```

Fig. 5: Setting Bluetooth

```
Meter2.tick(3000);
Serial.print(String(Meter1.getCurrentFlowrate()));
Serial.print("l/m");
Serial.print(":");
Serial.print(String(Meter1.getTotalVolume()));
Serial.print("L total");
Serial.print(":");
Serial.print(String(Meter2.getCurrentFlowrate()));
Serial.print("l/m");
Serial.print(":");
Serial.print(String(Meter2.getTotalVolume()));
Serial.print("L total");
Serial.println(" ");

oled.clearDisplay();
oled.setTextSize(1);
oled.setTextColor(1);
oled.setCursor(0, 0);
oled.println("Flow rate");
oled.setTextSize(2);
oled.setCursor(10, 12);
oled.println(Meter2.getCurrentFlowrate());
oled.setTextSize(2);
oled.setCursor(80, 12);
oled.println("l/m");
```

Fig. 6: Printing data to OLED display and sending data to Bluetooth module

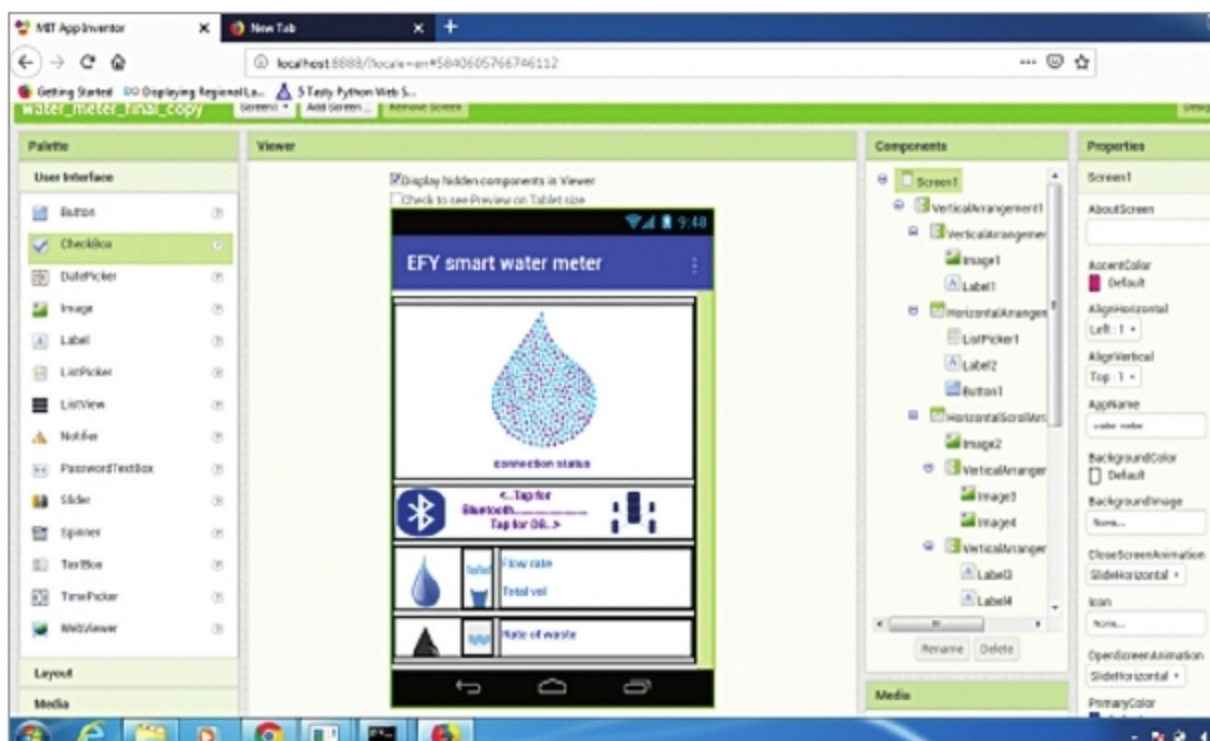


Fig. 7: Adding components in app layout

driver for the sensor as shown in Fig. 4.

Here, we set up the Bluetooth with baud rate at 9600. If this baud rate does not work, try default baud rate at 4800.

Now, set the data output that will be sent to Bluetooth with Serial.print (data to send) function as shown in Fig. 6.

And at last, set OLED display with oled.println (data to send) function.

We have finished the coding part; now let's start making an app using MIT App Inventor.

## Android app development

Open MIT App Inventor, make a layout and add the components as shown in Fig. 7.

Main components added in the layout are:

- 1 x Button for database
- 5 x Text views for Bluetooth connection status, inlet water flow rate, the total volume of water consumed,

wastewater flow rate, and total volume of water waste.

- TinyWebDB
- List picker for Bluetooth
- Bluetooth client
- Timer clock

The final app layout is shown in Fig. 8.

Now, go to MIT Code block and set the code as shown in Fig. 9. Save this code as smart\_meter.aia and generate its .apk file. This .apk file needs to be installed on your Android smartphone.

Next, we need to upload the Arduino code to Arduino Pro Mini.

**Note.** Arduino Pro Mini was used in the prototype to make the project small and easy; you can also use Arduino Uno for this project. Code and connections remain the same for both.

## Uploading code to Arduino Pro Mini

First, remove ATmega328 microcontroller from Arduino Uno board as shown in Fig. 10.

Connect Arduino Pro Mini with Arduino Uno as per Table 1.

Now, go to Tools and select Board as Arduino Pro Mini and upload the code (water\_meter.ino) to Arduino Pro Mini as shown in Fig. 11.

Next, we need to connect all the components to make the final smart water meter project.

The Arduino Pro Mini pin connec-

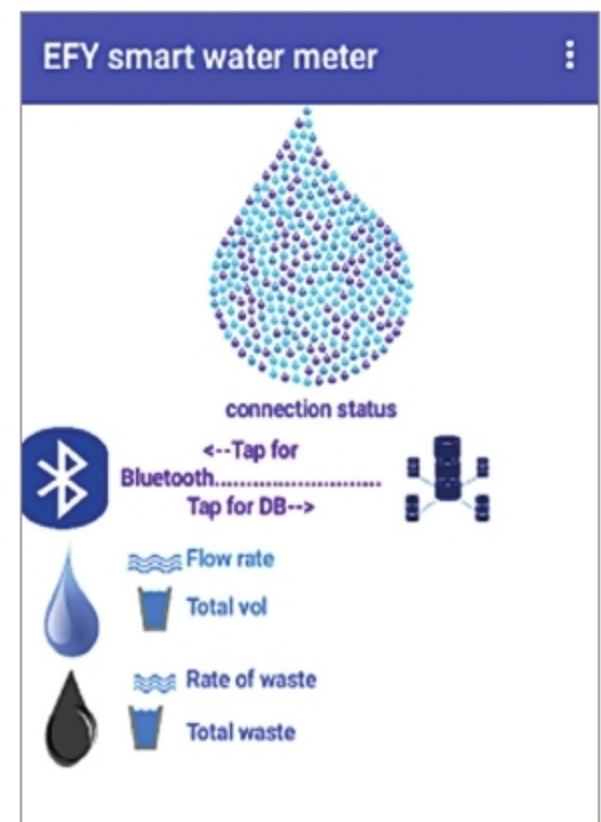


Fig. 8: App layout